Name: Syed Saud Ali (P2209111)
Class: EL/EP0611/01
Question: How does air quality affects a persons respiratory or health?
globalHealth dataset: https://www.kaggle.com/datasets/malaiarasugraj/global-health-statistics
airPollution dataset: https://www.kaggle.com/datasets/sazidthe1/global-air-pollution-data

Before answering the question, let's first read the datasets to be used as a dataframe in order to perform data analysis. I have imported 2 datasets, the airPollution which shows aqi(air quality index) value and more by country, and globalHealth which shows prevalence rate, mortality rate, incidence rate and more by country. I have also filterd the globalHealth dataset to show the last 5 years and only respiratory and cardiovascular disease categories to better fit our analysis. After filtering the globalHealth dataset, it is merged together with the airPollution dataset in order to do permuation testing.

In [1]:
```
import pandas as pd;
import matplotlib.pyplot as plt;
import numpy as np;
### Random Seed Generator
np.random.seed(42);
### Read csv files
airPollution = pd.read_csv('global_air_pollution_data.csv');
globalHealth = pd.read_csv('Global_Health_Statistics.csv');
### Filter globalHealth from 2020 to 2024
globalHealth2024 = globalHealth[(globalHealth['Year']>=2020) & (globalHea
### Filter globalHealth to only show disease categoty of Respiratory and
globalHealth2024CardioRespit = globalHealth2024[(globalHealth2024['Diseas
### Merge filtered globalHealth with airPollution
globalMergePollution = pd.merge(airPollution, globalHealth2024CardioRespi
```

In order to use permutation method, we should first form a hypothesis, which is stated here: there is no significant difference in disease prevalence rate between high pollution and low pollution regions. Let's now do permutation method.

In [2]:
```
globalMergePollutionSample = globalMergePollution.sample(10000,random_sta
aqi, prevalenceRate = globalMergePollutionSample['aqi_value'],globalMerge
### Find aqi median to separate high and low pollution regions
aqiMedian = np.median(aqi);
### Split aqi values into high and low pollution groups
highPollution = prevalenceRate[aqi>=aqiMedian];
lowPollution = prevalenceRate[aqi<aqiMedian];
### Compute observed difference by finding difference of mean
obsMeanDiff = np.mean(highPollution)-np.mean(lowPollution);
print(f"Observed mean: {obsMeanDiff}");
```

```
Observed mean: -0.12551004867301785
```

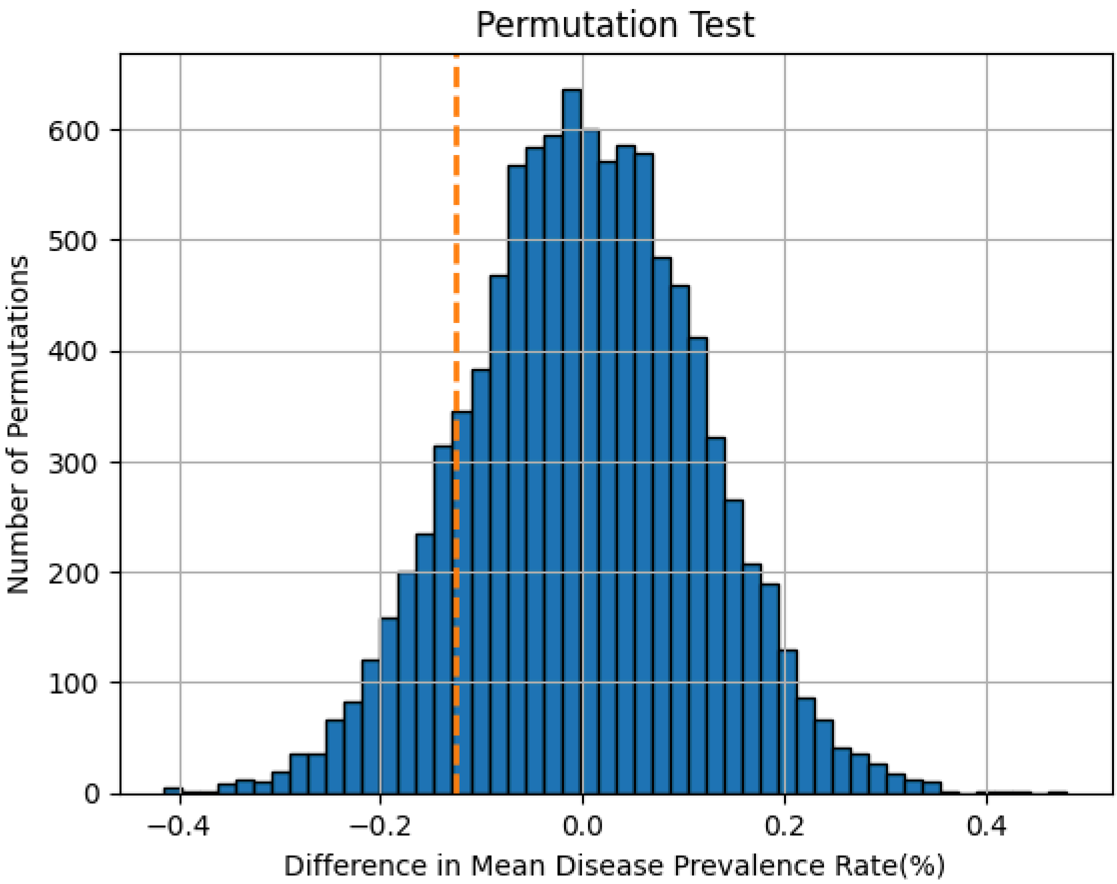
Before we perform permutation tests, we should first define 2 functions that helps to find the difference between the permutation sample gathered.

In [3]:
```
### Generate Purmutation Sample
def permutation_sample(data1, data2):
    '''Generates a permutation sample from two data sets'''

    # Concatenate the data sets and shuffle data points
    all_data = np.concatenate((data1, data2));

    # Permute the concatenated data
    permuted_data = np.random.permutation(all_data);

    # Assign new labels to the data points based on their permuted positi
    perm_sample_1 = permuted_data[:len(data1)];
    perm_sample_2 = permuted_data[len(data1):];

    return perm_sample_1, perm_sample_2;

def draw_perm_diff(data1, data2, num_reps = 1):
    '''Obtains permutation test distribution for the difference in mean o

    # Initialize vector to hold replicates
    perm_diff_reps = np.zeros(num_reps);

    # Generate replicates
    for i in range(num_reps):
        ### gather permutation samples
        perm_sample_1, perm_sample_2 = permutation_sample(data1,data2);
        ### Find Difference of Mean between 2 samples
        perm_diff_reps[i] = np.mean(perm_sample_1) - np.mean(perm_sample_

    return perm_diff_reps;
```

Once the functions are set, we can finally start to perform permutation test.

In [4]:
```
### Perform Permutated Test
permMeanDiffs = draw_perm_diff(highPollution, lowPollution, num_reps=1000
### Compute p-value
pVal = np.mean(np.abs(permMeanDiffs)>=np.abs(obsMeanDiff));
### Compute confidence intervals
confInt = np.percentile(permMeanDiffs,[2.5,97.5]);
print(f"95% Confidence interval: {confInt}");
### Output Results
print(f"P-value: {pVal:.4f}");
### Plot Histogram
_ = plt.hist(permMeanDiffs,bins=50,color='C0',ec='black');
_ = plt.xlabel("Difference in Mean Disease Prevalence Rate(%)");
_ = plt.ylabel("Number of Permutations");
_ = plt.title("Permutation Test");
_ = plt.axvline(x=obsMeanDiff,color='C1',linestyle='--',linewidth=2);
plt.grid(True);
plt.show();
```

```
95% Confidence interval: [-0.2232125   0.22135557]
P-value: 0.2723
```



Once the results from the permutation method is gathered, we can that the observed correlation between AQI value and prevalence rate of diseases is not statistically significant, as the p-value is significantly larger than 0.05. Thus, our test strongly supports our hypothesis that there is no relationship between air quality and prevalence rate of diseases. This is further reinforced by the 95% condifence interval as both bound are close to 0, showing that there the difference in disease prevalence rate between high and low aqi regions are negligible.
Moreover, as seen from the histogram, we can observe the dashed line is more to the right side, we can observe that regions with a higher aqi value surprisingly has lower disease prevalence rate compared to regions with lower aqi value. This counterintuitive result suggest that other factors such as habits of population in different regions, the quality of healthcare or environmental differences influence prevalence rate of diseases beyond air quality.